

# QuickDraw — trained bases vs classical baselines

## 1. Main result

We introduce a parametric quantum-circuit family of unitary image bases with  $O(N \log N)$  forward / inverse cost and  $O(\log N)$  trainable parameters per dimension. **(i)** On line-drawing imagery (QuickDraw,  $32 \times 32$ ), the trained block-wrapped basis exceeds every linear classical baseline tested — including the dataset-fitted optimum BlockPCA-8 — by **+5.97 dB** PSNR over BlockDCT-8 at  $\rho = 0.20$ , after  $\approx 5$  minutes of training on a single RTX 3090. **(ii)** On natural images (DIV2K,  $256 \times 256$ ), the same

family matches BlockDCT-8 to within  $\approx 0.3$  dB, saturating the **Ahmed–Natarajan–Rao limit** under which the KLT of a stationary Gaussian AR(1) source converges to the DCT. **(iii)** Accordingly, the parametric basis offers a rigorous improvement over BlockDCT on image distributions whose patch statistics depart from stationary Gaussian AR(1), and parity on those that do not.

## 2. Setup + pipeline

$$m = n = 5, \quad d = 2^m \cdot 2^n = 32 \times 32 = 1024$$

$$N_{\text{train}} = 500, \quad N_{\text{test}} = 50, \quad \text{seed} = 42, \quad \rho \in \{0.05, 0.10, 0.15, 0.20\}$$

Per image  $x \in [0, 1]^d$  with basis  $\Phi$ :

$$y = \Phi x \quad (\text{forward})$$

$$k = \lfloor \rho \cdot d \rfloor, \quad y'_i = \begin{cases} y_i & \text{if } |y_i| \in \text{top-}k \\ 0 & \text{else} \end{cases} \quad (\text{compression})$$

$$\hat{x} = \text{clip}(\Phi^{-1}y', 0, 1), \quad \text{MSE} = \frac{1}{d} \|x - \hat{x}\|_2^2, \quad \text{PSNR} = 10 \log_{10}(1/\text{MSE})$$

For block transforms: same pipeline applied independently to each  $8 \times 8$  tile, top- $k$  pooled across all blocks.

**AR(1) (first-order autoregressive).** Each pixel =  $\rho_{\text{AR}} \times \text{previous} + \text{small Gaussian noise}$ :

$$x_n = \rho_{\text{AR}} x_{n-1} + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

$\rho_{\text{AR}} \rightarrow 1$  Gaussian  $\rightarrow$  KLT collapses to DCT (Ahmed–Natarajan–Rao). Empirical lag-1 autocorrelation  $\hat{\rho}_{\text{AR}} = \frac{1}{2}(\hat{\rho}_{\text{row}} + \hat{\rho}_{\text{col}})$  across multiple samples: **DIV2K natural images** (4 different scenes, centre  $256 \times 256$  crops) cluster in  $\hat{\rho}_{\text{AR}} \in [0.84, 0.99]$  — close to the image-row-like regime,  $\rho \approx 1$ , where BlockDCT is near-optimal; **QuickDraw drawings** (3 samples, single  $28 \times 28$  each) cluster tightly at  $\hat{\rho}_{\text{AR}} \approx 0.70$  — further from the AR(1)–Gaussian limit, leaving room for a trained basis to beat BlockDCT. The two distributions barely overlap, exactly the QuickDraw vs DIV2K asymmetry the results table shows. Distinct from the keep-ratio  $\rho$  above.

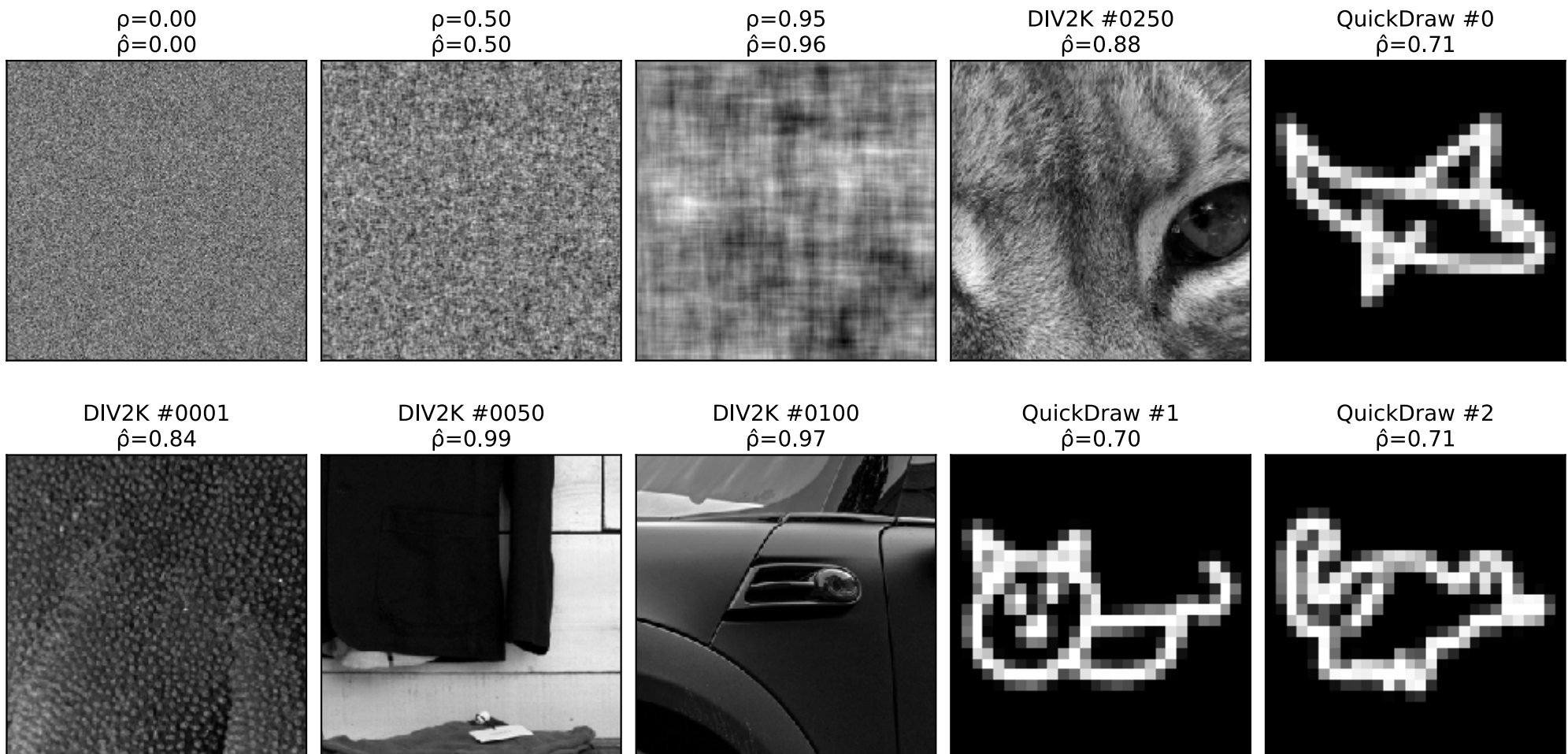


Figure 1: **Top row:** three synthetic AR(1)–Gaussian fields at  $\rho = 0, 0.5, 0.95$  (sanity-checking  $\hat{\rho}$ ); DIV2K 250 centre crop; QuickDraw drawing 0 **Bottom row:** three more DIV2K samples (1, 50, 100) and two more QuickDraw drawings (1, 2) — confirming the  $\hat{\rho}_{\text{AR}}$  values are robust across samples and the two datasets cluster at different points on the AR(1) axis. QuickDraw panels are single  $28 \times 28$  drawings nearest-neighbour upscaled  $9 \times$  for visual size;  $\hat{\rho}$  is computed on the native pixels.

**3. Fits — what is  $\Phi$  for each method?**

| method                                  | $\Phi$ source                                                                                                                                                                                                                                                                                                                        | $\Phi$ shape                                                                                                                                                                                  | forward $y =$                                                           | compression $y' =$                                                                                                                                           | inverse $x =$                                    |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| bd_pca (bilateral 2D-PCA)               | $U, V$ from SVDs of column- and row-stacked centered images:<br>$\Sigma_{\text{col}} = \frac{1}{NW-1} \sum_i (X_i -  (X)) (X_i -  (X))^T (H \times H)$<br>$\Sigma_{\text{row}} = \frac{1}{NH-1} \sum_i (X_i -  (X))^T (X_i -  (X)) (W \times W)$<br>fit on $N = 500$ images at $H = W = 32$<br>( $NH = NW = 16000$ samples per axis) | $U$ is $H \times H = 32 \times 32$ , full rank<br>$V$ is $W \times W = 32 \times 32$ , full rank<br>(each axis has $16000 \gg 32$ samples)                                                    | $Y = U^T(X -  (X))V$ ,<br>shape $H \times W$                            | top- $k$ on $Y$ : $Y_{ij} \cdot \mathbb{1}[ Y_{ij}  \geq \tau_k]$ , $k = \lfloor \rho \cdot HW \rfloor = \lfloor \rho \cdot d \rfloor$<br>(same rate as DCT) | $UY'V^T +  (X)$                                  |
| block_bd_pca_8 (block bilateral 2D-PCA) | $U_b, V_b$ are separable column + row eigenbases at the $8 \times 8$ patch level, fit on $N_p = 8000$ pooled patches<br>(500 images $\times 4 \times 4 = 16$ blocks per image),<br>via SVDs on $N_p b \approx 64000$ samples per axis                                                                                                | $U_b, V_b$ are each $b \times b = 8 \times 8$ , full rank, shared across all $4 \times 4$ blocks.<br>Separable constraint: 128 params total vs $b^2 \times b^2 = 4096$ for unconstrained KLT. | per block: $Y_b = U_b^T(P -  (P))V_b$<br>(shape $8 \times 8$ per block) | top- $k$ pooled across all blocks, keep $k = \lfloor \rho \cdot 1024 \rfloor$ largest entries of the $4 \times 4 \times 8 \times 8$ tensor                   | per block: $U_b Y_b V_b^T +  (P)$ , then re-tile |
| dct (global)                            | closed-form (DCT-II):<br>$\Phi[k, n] = \alpha_k \cos(\pi(n + 1/2)k/N)$<br>$\alpha_0 = \sqrt{\frac{1}{N}}$ , $\alpha_{k \geq 1} = \sqrt{\frac{2}{N}}$<br>$N = 32$ , no fit, no mean                                                                                                                                                   | $d \times d$ analytically                                                                                                                                                                     | $\Phi x$ (no mean)                                                      | top- $k$ : $y_i \cdot \mathbb{1}[ y_i  \geq \tau_k]$<br>(rank: keep $i = 0, \dots, k - 1$ )                                                                  | $\Phi^T y'$                                      |
| block_dct_8                             | closed-form (DCT-II) at $N = 8$ :<br>$\Phi[k, n] = \alpha_k \cos(\pi(n + 1/2)k/8)$<br>shared across all blocks                                                                                                                                                                                                                       | $64 \times 64$ per block analytically<br>tiled identically across the image                                                                                                                   | per block: $\Phi p$                                                     | top- $k$ pooled across all blocks<br>(rank: per-block keep $i = 0, \dots, k_b - 1$ , $k_b = \lfloor \rho \cdot 64 \rfloor$ )                                 | per block: $\Phi^T y'$ , then re-tile            |

Table 1: Concrete definition of  $\Phi$  and the full pipeline (forward  $\rightarrow$  compression  $\rightarrow$  inverse) for each method. All four  $\Phi$  are **orthogonal** ( $\Phi\Phi^T = I$ ), so inverse = transpose. 2D = separable (apply 1D transform along rows then columns).

**Top- $k$  rule** (default, headline results): keep the  $k$  coefficients with largest **[magnitude]**, set the rest to zero — pooled across all blocks for block transforms.

**Rank rule** (in parentheses; used by the `_rank` control variants): keep the **first**  $k$  coefficients in the basis’s natural ordering. For **PCA** this means the  $k$  **highest eigenvalue** directions (rows of  $V^T$  are sorted by descending  $\lambda$ ); for **DCT** this means the  $k$  **lowest frequency** components (DC first, then increasing

### 3.1. Why DCT $\approx$ block PCA in the limit

$$\Phi_{\text{block\_pca}_8} = \text{KLT}(\Sigma_b) \longrightarrow \Phi_{\text{block\_dct}_8} = \text{KLT}\left(\lim_{\rho \rightarrow 1} \Sigma_{\text{AR}(1)}(\rho)\right)$$

DCT = what KLT becomes if the patch covariance is the analytic AR(1)–Gaussian limit. Empirically the two differ by only 0.68 dB on QuickDraw — natural patches are nearly AR(1)–Gaussian.

## 4. Results — PSNR (dB), seed=42

| unblocked (full $32 \times 32$ ) | $\rho = 0.05$ | $\rho = 0.10$ | $\rho = 0.15$ | $\rho = 0.20$ |
|----------------------------------|---------------|---------------|---------------|---------------|
| Trained PDFT bases (ours)        |               |               |               |               |
| ★ qft                            | <b>16.72</b>  | <b>19.58</b>  | <b>22.05</b>  | <b>24.35</b>  |
| ★ entangled_qft                  | <b>16.72</b>  | <b>19.58</b>  | <b>22.05</b>  | <b>24.35</b>  |
| ★ tebd                           | <b>16.72</b>  | <b>19.58</b>  | <b>22.06</b>  | <b>24.36</b>  |
| Classical, top- $k$ rule         |               |               |               |               |
| bd_pca                           | <b>18.63</b>  | <b>21.99</b>  | <b>24.87</b>  | <b>27.57</b>  |
| dct                              | 16.13         | 18.44         | 20.33         | 22.03         |
| fft                              | 15.26         | 17.32         | 19.01         | 20.56         |
| Classical, rank rule (control)   |               |               |               |               |
| dct_rank                         | 13.84         | 15.24         | 16.49         | 17.63         |

| 8×8 block-wrapped              | $\rho = 0.05$ | $\rho = 0.10$ | $\rho = 0.15$ | $\rho = 0.20$ |
|--------------------------------|---------------|---------------|---------------|---------------|
| Trained PDFT bases (ours)      |               |               |               |               |
| ★ rich                         | 18.79         | 23.78         | 28.27         | <b>32.69</b>  |
| ★ <b>real_rich</b>             | <b>18.80</b>  | <b>23.80</b>  | <b>28.30</b>  | <b>32.67</b>  |
| ★ blocked                      | 18.12         | 22.40         | 26.20         | 30.05         |
| Classical, top- $k$ rule       |               |               |               |               |
| block_dct_8                    | <b>17.20</b>  | <b>20.70</b>  | <b>23.72</b>  | <b>26.63</b>  |
| block_bd_pca_8                 | 17.00         | 20.50         | 23.41         | 26.05         |
| block_fft_8                    | 15.97         | 18.74         | 21.09         | 23.34         |
| Classical, rank rule (control) |               |               |               |               |
| block_dct_8_rank               | 13.50         | 14.81         | 15.78         | 16.97         |

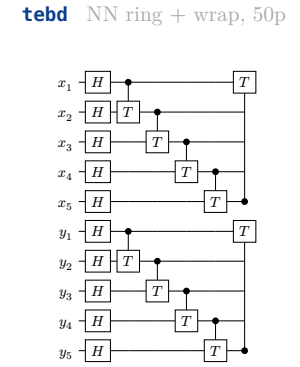
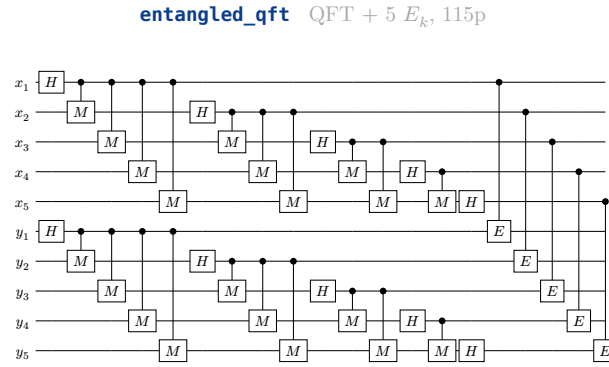
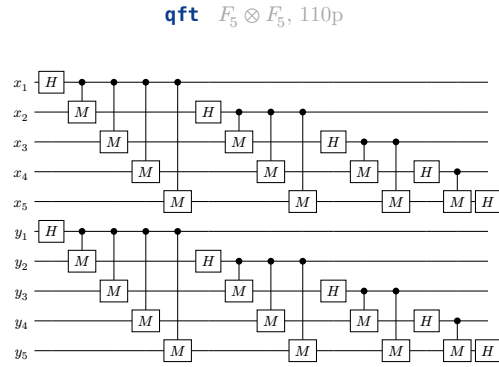
Table 2: Side-by-side: unblocked / full-image methods (left, gray header) vs  $8 \times 8$  block-wrapped methods (right, light blue header). Each column groups trained PDFT bases, classical top- $k$  baselines, and the rank-rule control. ★ = trained PDFT basis (this work); unmarked rows

## 5. classical baselines. Bold = best-in-group at each

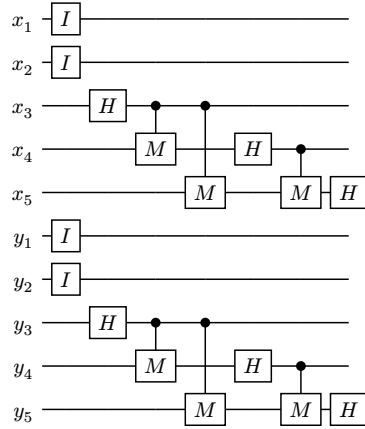
keep ratio. **real\_rich** ★ wins at three of the four ratios (red highlight) using **half** the parameters of rich; rich edges it out at  $\rho = 0.20$  by 0.02 dB. The two block-wrapped trained bases are essentially tied within single-seed noise; the headline emphasises **real\_rich** for the parameter-count advantage. **mera** omitted (incompatible at  $m + n = 10$ ).

### 5.1. Tensor networks of the trained bases at $m = n = 5$

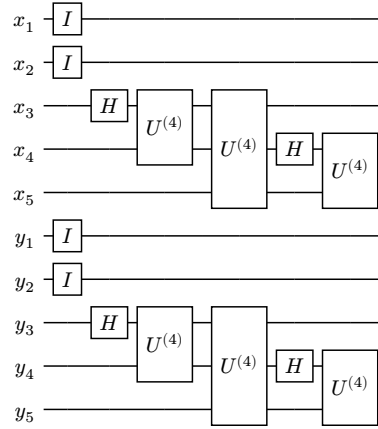
10-qubit layout (rows  $x_1, \dots, x_5$  top, cols  $y_1, \dots, y_5$  bottom). Seven sub-circuits arranged in a 4-column grid for side-by-side comparison. All separable except **entangled\_qft**.



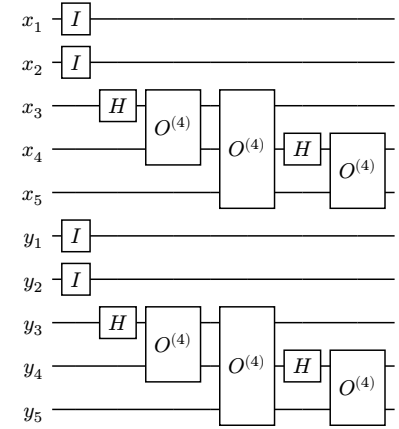
**blocked**  $I_4 \otimes F_3$  per dim, 30p



**rich**  $I_4 \otimes U^{\text{inner}}$ , 108p



**real\_rich**  $I_4 \otimes O^{\text{inner}}$ , 42p (HEADLINE)



Listing 1: Six trained-basis topologies at QuickDraw geometry ( $m = n = 5$ , 10 qubits per circuit). Top row — unblocked: **qft**, **entangled\_qft**, **tebd** (all at the same scale and row-spacing so their 10-wire heights match exactly). Bottom row — block-wrapped: **blocked**, **rich**, **real\_rich** (headline). **mera** is structurally inapplicable here ( $m + n = 10 \neq 2^p$ ) and omitted. Param counts cover both row and col registers.

## 5.2. Geometry comparison at a glance

| basis         | topology                     | row-col coupling | params/dim | PSNR at $\rho = 0.20$ |
|---------------|------------------------------|------------------|------------|-----------------------|
| qft           | all-to-all per dim           | none             | 55         | 24.35                 |
| entangled_qft | all-to-all + 5 cross         | yes ( $E_k$ )    | 60         | 24.35                 |
| tebd          | NN ring per dim              | none             | 25         | 24.36                 |
| blocked       | outer- $I$ + inner QFT-3     | none             | 15         | 30.05                 |
| rich          | outer- $I$ + inner $U^{(4)}$ | none             | 54         | 32.69                 |
| real_rich     | outer- $I$ + inner $O^{(4)}$ | none             | 21         | 32.67                 |

Table 3: All bases at QuickDraw geometry  $m = n = 5$ . Param counts are per dim (multiply by 2 for full transform unless row/col are tied). PSNR from Table earlier in this doc.

## 6. Training loss curves

**Loss definition.** Per-image loss is

$$\ell(\theta, x) = \sum_{i,j} \left| x_{ij} - \left( T_{\theta}^{\dagger} \text{top}_K(T_{\theta} x) \right)_{ij} \right|^2, \quad K = 102 \ (\rho = 0.10),$$

summed over all 1024 pixels of a  $32 \times 32$  image, where  $T_{\theta}$  is the parametric forward unitary and  $T_{\theta}^{\dagger}$  its inverse (conjugate transpose). The plotted y-axis is this loss averaged over the 75-image validation split.  $K = 102$  is the **training** keep-count ( $\rho = 0.10$ , fixed across the run); the headline Table 2 reports per-image PSNR at  $\rho \in \{0.05, 0.10, 0.15, 0.20\}$  on a separate 50-image test split, so the absolute scales here do not equal the dB values there — only the ranking is shared.



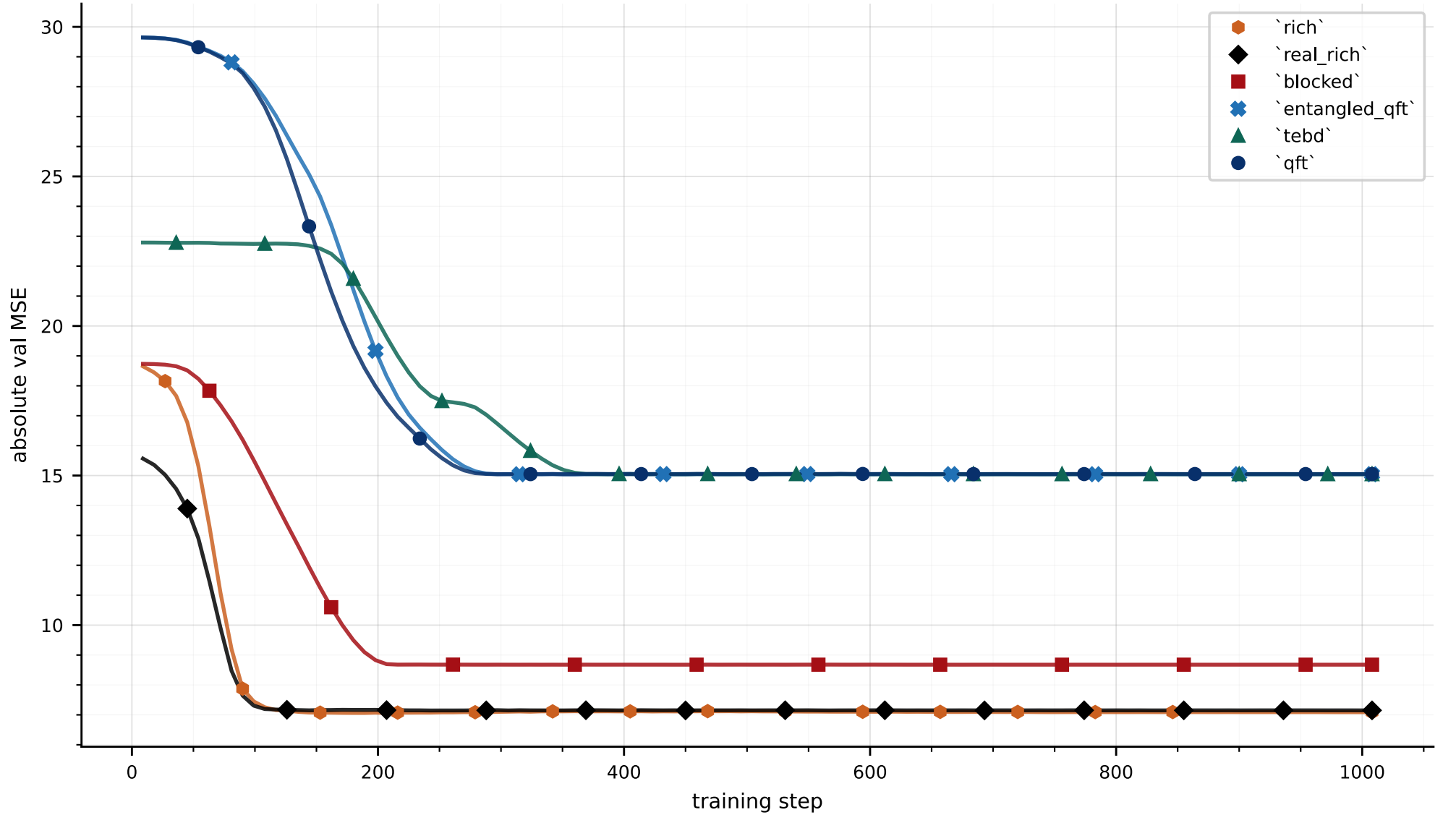


Figure 2: Per-epoch validation MSE for each trained basis on QuickDraw at the headline 1008-step training budget. All six bases share a single linear y-axis: absolute MSE, which is the operational quantity ( $\max \text{PSNR} \equiv \min \text{MSE}$ ), so the right-edge curve ordering reproduces the headline PSNR ranking by inspection. The block-wrapped bases (`real_rich`, `rich`, `blocked`) start at lower  $L_0$  than the unblocked ones because block transforms are already sensible bases at random init for  $32 \times 32$  line drawings. Each basis is a (colour, marker) pair from a colourblind-safe palette; coincident curve pairs (`qft`  $\equiv$  `entangled_qft`) overlap exactly because their parametric families compile to the same operator at this circuit depth. `mera` is omitted because  $m + n = 10$  is not a power of 2 and the MERA hierarchy is structurally inapplicable. Loss is essentially flat past step  $\approx 300$  for every basis.

## 6.1. Appendix: 2000-step plateau check

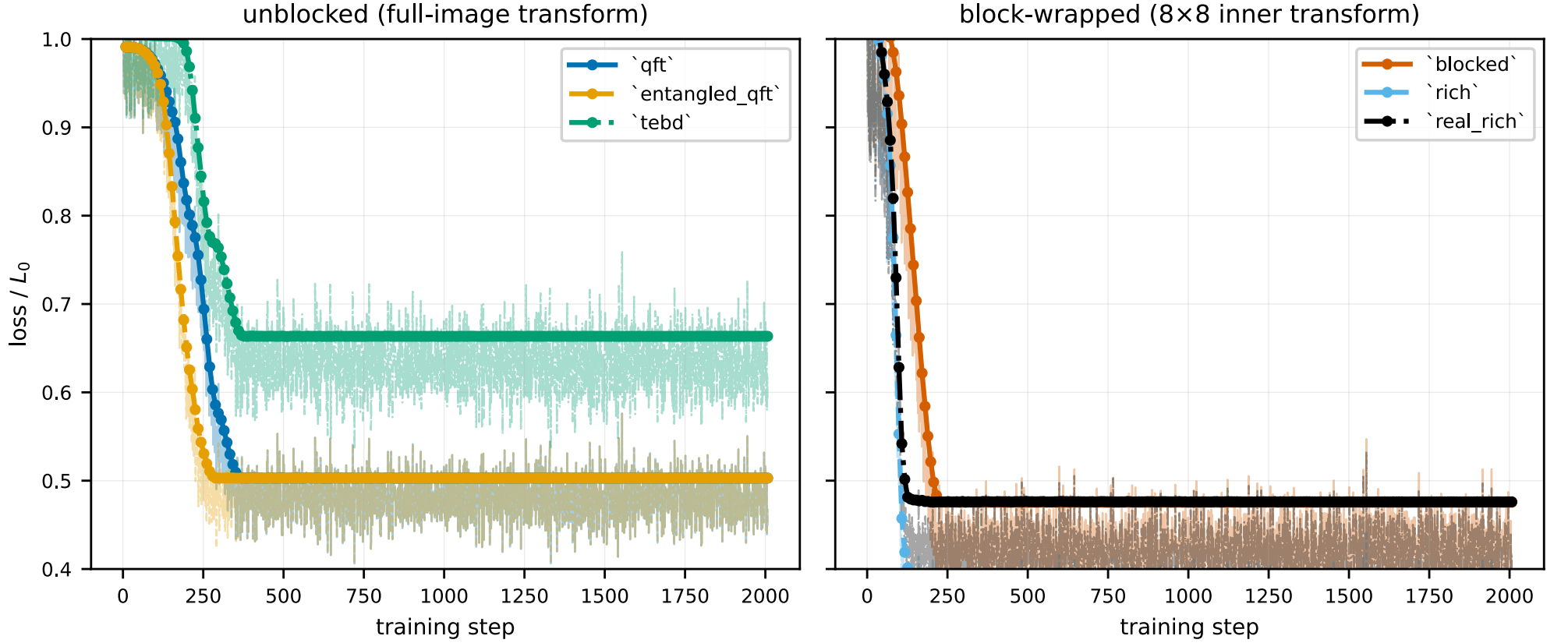


Figure 3: Same plot as the headline figure but trained for  $\approx 2000$  steps (epochs = 223) instead of  $\approx 1000$  (epochs = 112). Verifies that QuickDraw losses have plateaued by the headline budget — the windowed-mean training loss between steps 1000 and 2000 moves by  $\leq 0.01$  dB equivalent. Headline-table values shift by  $\leq 0.05$  dB between the 1000-step and 2000-step runs (*real\_rich* for example:  $32.67 \rightarrow 32.62$  at  $\rho = 0.20$ ); the shifts are below the single-seed noise floor on  $n_{\text{test}} = 50$ .

## 7. Reconstructions — 2 representative images $\times$ keep ratios $\times$ bases

### 7.1. Image 0 — cat sketch

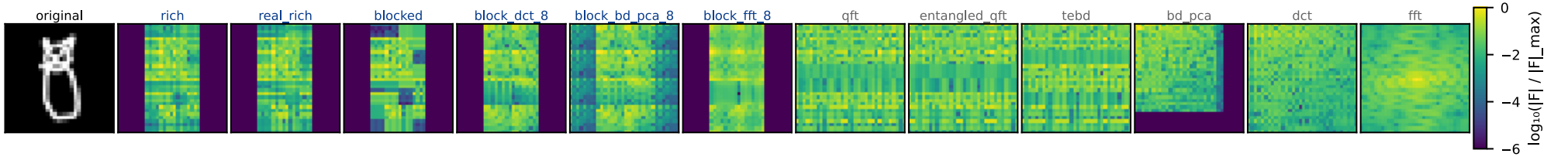


Figure 4: Frequency-space spectra (peak-normalised  $\log|F|$ , viridis, shared color scale) for image 0 under each basis. Same column order as the recon grid below. Note the trained block-wrapped bases (blue headers) push energy into a small number of low-coefficient cells per block — visible as the bright clusters in `rich`/`real_rich`/`blocked` and the band structure in `block_dct_8`/`block_bd_pca_8`. The `bd_pca` panel shows energy concentrated in the top-left corner (low-frequency / high-eigenvalue) along both axes — the separable-KLT analog of the DCT spectrum.

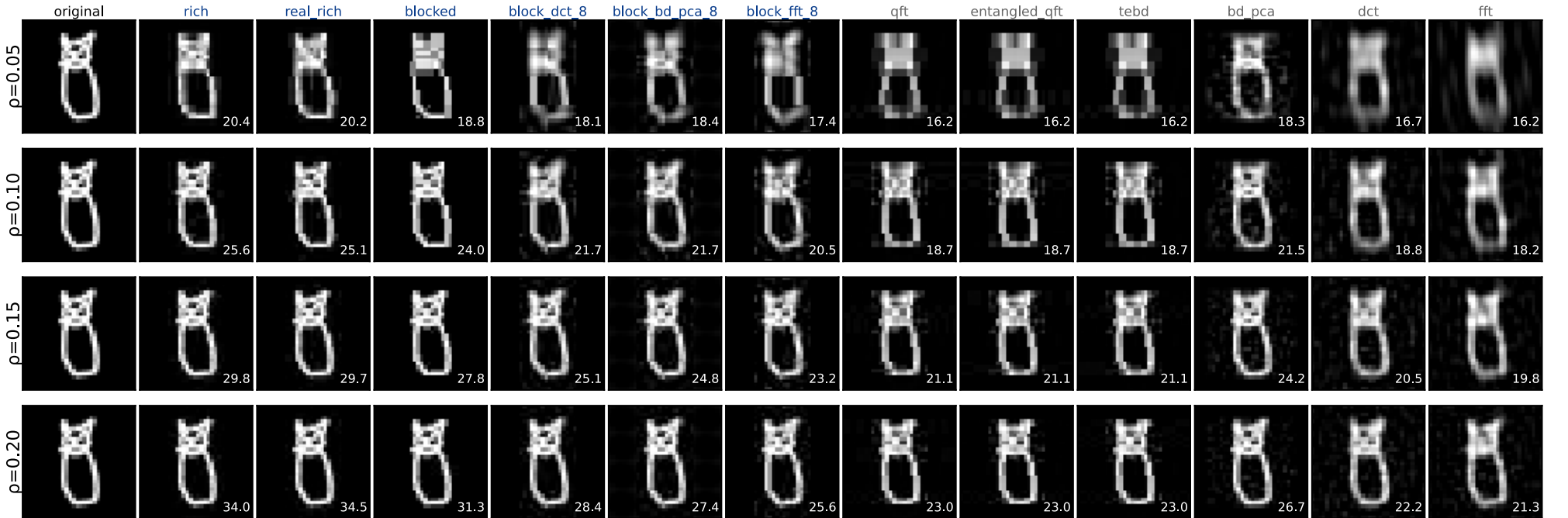


Figure 5: QuickDraw test image 0 reconstructed at four keep ratios  $\rho \in \{0.05, 0.10, 0.15, 0.20\}$  (rows). Same column order as the freq-space figure above. PSNR (dB) annotated bottom-right of each cell. At  $\rho = 0.20$ : trained `rich`  $\approx 34$  dB vs `block_dct_8`  $\approx 28$  dB (+6 dB) vs `dct`  $\approx 22$  dB (+12 dB).

## 7.2. Image 2 — fish silhouette



Figure 6: QuickDraw test image 2 reconstructed at the same four keep ratios. Same column layout as image 1 This image is among the strongest cases for trained tensor-network bases: at  $\rho = 0.20$ , trained rich  $\approx 44$  dB vs block\_dct\_8  $\approx 32$  dB (+12 dB) vs dct  $\approx 25$  dB (+19 dB). Even at  $\rho = 0.05$  (top row) rich and real\_rich already preserve the recognisable fish-stroke topology while dct and fft collapse to a blurred blob.

## 8. Matching summary — bench name $\rightarrow$ paper figure

| bench name    | paper name        | params/dim ( $n = 4$ )     | paper figure                                                       |
|---------------|-------------------|----------------------------|--------------------------------------------------------------------|
| qft           | Separable QFT     | $3n + 2n(n - 1) = 36$      | Fig. 1d, Tab. 1                                                    |
| entangled_qft | Entangled QFT     | $36 + n = 40$              | Fig. 3, Eq. (entangled_qft)                                        |
| tebd          | TEBD ring         | $3n + 2n = 20$             | App. C left, Fig. (tebd_circuit)                                   |
| mera          | MERA hierarchical | $3n + 4(n - 1) = 24$       | App. C right, Fig. (mera_circuit)                                  |
| blocked       | Blocked QFT       | 15 ( $m_{\text{in}} = 3$ ) | <code>\\$block_wrapper</code> , Fig. (block_basis_circuits) bottom |
| rich          | RichBasis         | 54 ( $m_{\text{in}} = 3$ ) | Fig. (block_basis_circuits) middle                                 |
| real_rich     | RealRichBasis     | 21 ( $m_{\text{in}} = 3$ ) | Fig. (block_basis_circuits) top — headline                         |

Table 4: Mapping from `pdft_benchmarks` basis name to the paper’s naming and circuit figure. Param counts use the conventions in `main.tex` `\$sec:qft_basis` and `\$sec:block_wrapper`.